Oddech

A robot that breathes the room with you.

Living airway cells tell it **how harmful** the air is, its map tells it **where**, and it tells you **what to do**.

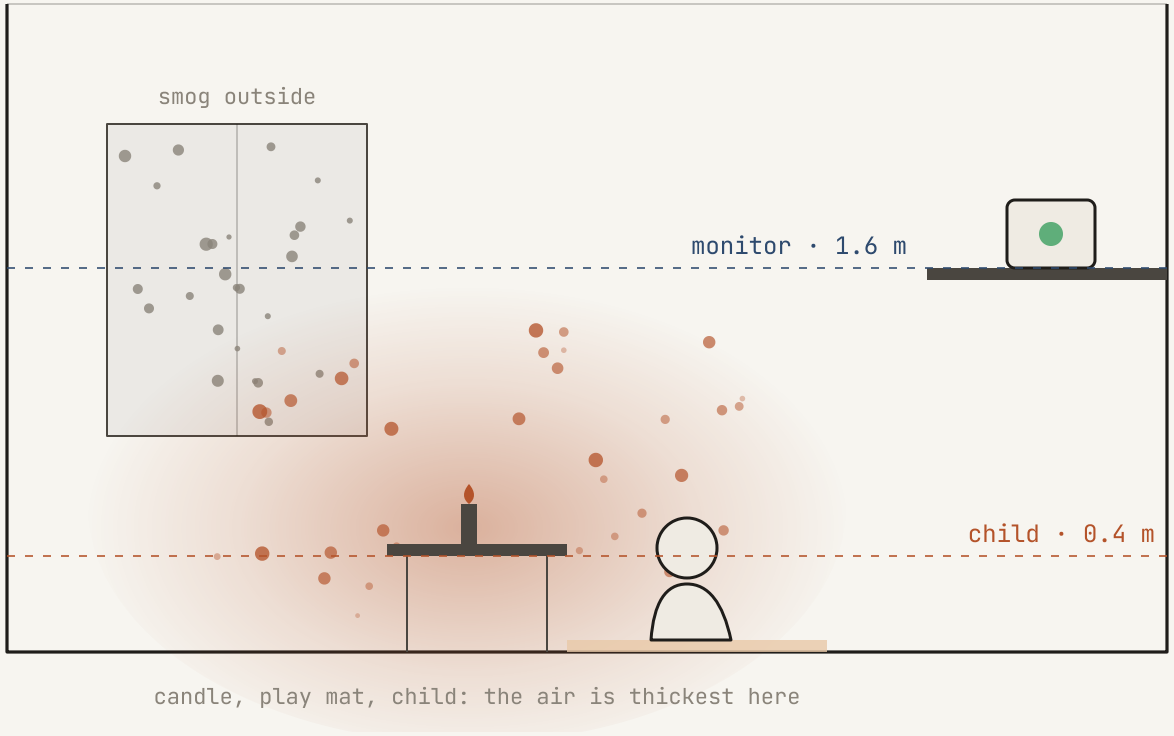
Julian Evan Chrisnanto · Antonio Galiza Cerdeira Gonzalez · Xiyue (Thea) Cao · Aleksandra Skrzypiec



Oddech, Polish for “breath” · rendered from the robot's 3D model

Oddech

Kraków's smog comes indoors. *The shelf monitor misses the child.*



candle, play mat, child: the air is thickest here

87 %

of our time is spent indoors, with one monitor on a shelf.

6 to 20 ×

dirtier air right next to a source than in a well-mixed room.

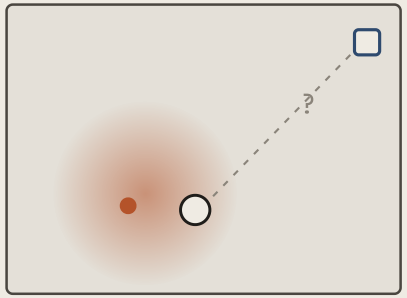
Not yet

Kraków's ban on coal heating cleaned the air, but the EU limit is still not met.

Sources: Klepeis et al. 2001; Acevedo-Bolton et al. 2011; Kicińska & Jawdyńska 2025

Oddech

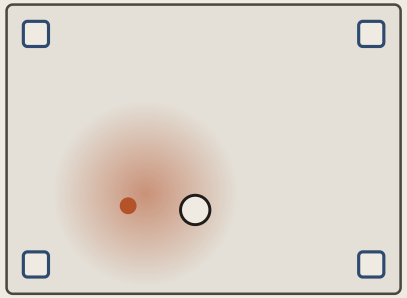
More devices, more numbers. *Still nobody asks the body.*



One monitor on a shelf

One number, far from the child, at the wrong height.

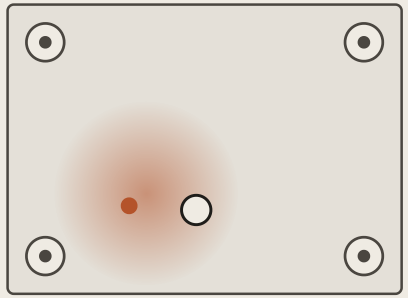
- ○ Reads the air the child breathes
- ○ Knows how harmful it is
- ○ Tells you, and acts



A sensor in every corner

More numbers, still not where the child plays; units drift apart.

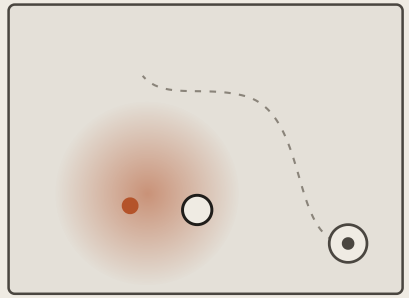
- ◑ Reads the air the child breathes
- ○ Knows how harmful it is
- ○ Tells you, and acts



A purifier in every corner

Cleans where it stands, all day, without knowing what the child breathes.

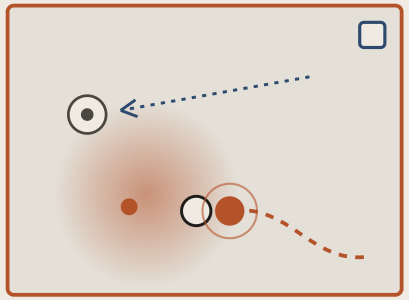
- ○ Reads the air the child breathes
- ○ Knows how harmful it is
- ◑ Tells you, and acts



A purifier robot

Moves a filter around; measures at its own body, not the harm.

- ◑ Reads the air the child breathes
- ○ Knows how harmful it is
- ◑ Tells you, and acts



Oddech

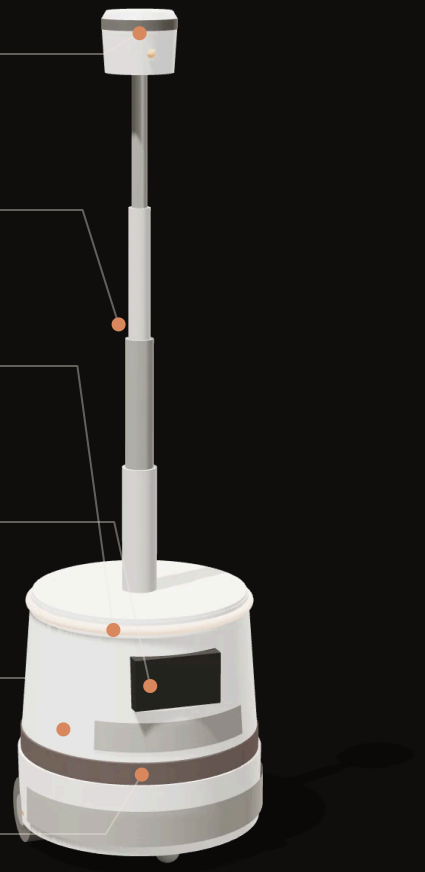
Goes to the child, asks living cells how harmful the air is, then tells and acts.

- ● Reads the air the child breathes
- ● Knows how harmful it is
- ● Tells you, and acts

Every setup adds devices. *None says how harmful the air is where the child breathes.*

Our assessment · units drift apart: Castell et al. 2017 · purifier robot: Ecovacs AIRBOT Z1

Oddech: one robot that *breathes the room with you.*

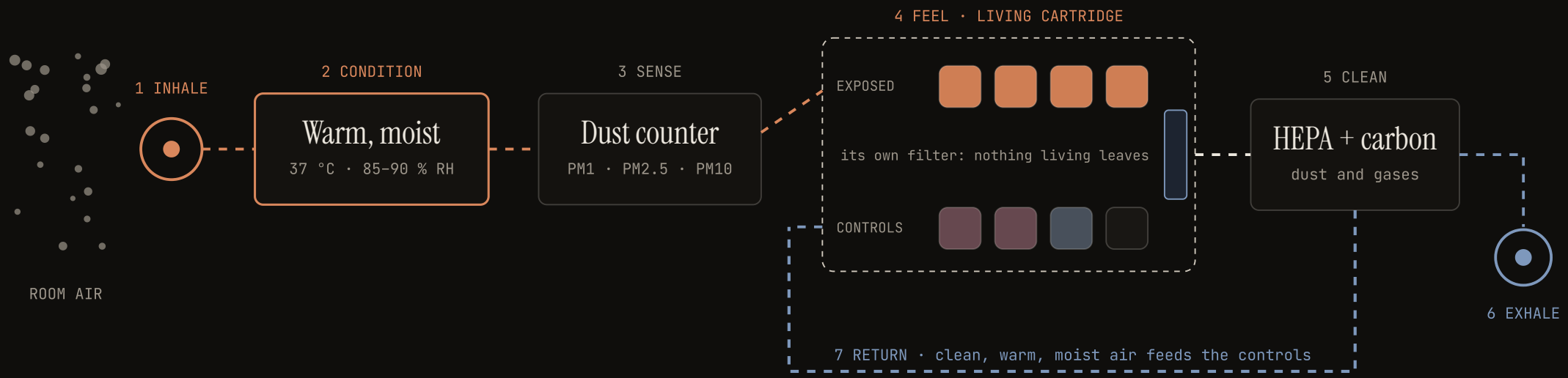


01	Breathe	at the child's, a seated and a standing height
02	Feel	living airway cells glow when the air harms them
03	Map	a live map of the room, and where it is unsure
04	Tell	plain words: what, where, how sure, what to do
05	Act	the purifier, the window, the phone

Design targets for a robot not yet built · 3D model from our Gazebo package

Oddech

Every ten minutes, *one breath.*



Warm and moist

37 °C and 85–90 % RH: the conditions a mobile field system keeps its cells in.

Dry air harms too

Low humidity alone hurts airway cells, so the bay controls it.

Fair controls

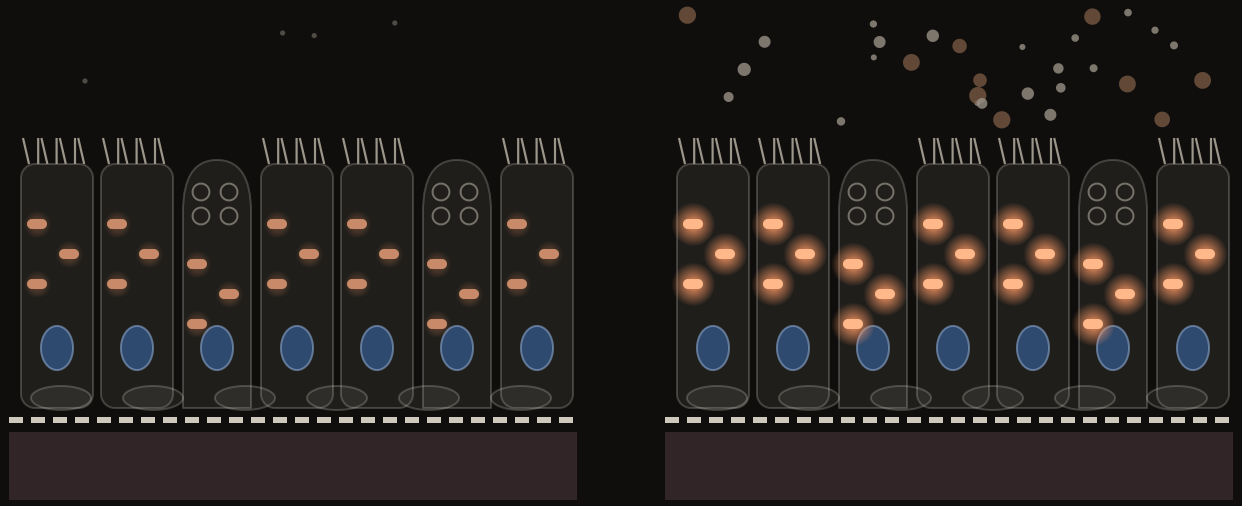
The control wells breathe the robot's own clean, conditioned air.

Nothing living leaves

The cartridge has its own filter; HEPA and carbon clean the rest.

Oddech

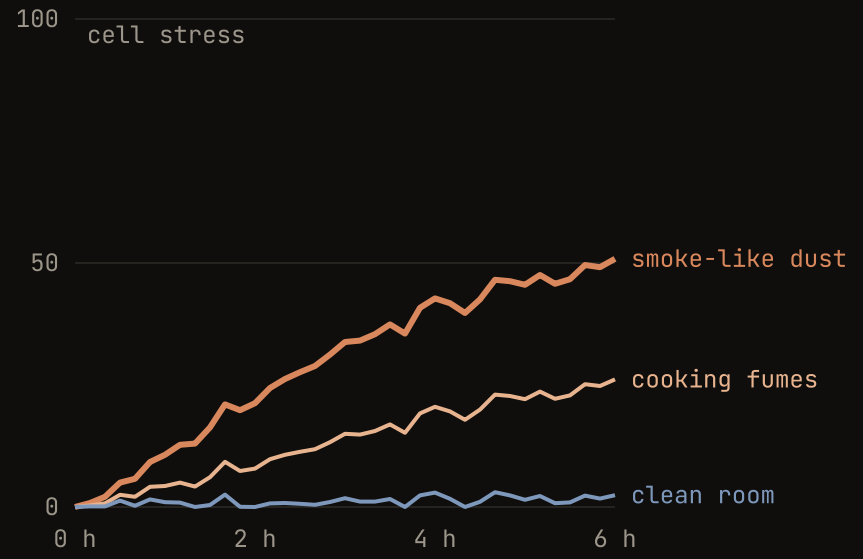
Living airway cells *feel what a dust counter cannot.*



FILTERED CONTROL
same cells, the same air minus the dust

ROOM AIR
the dust stresses the cells: they glow

ciliated · goblet · basal cells · membrane · medium

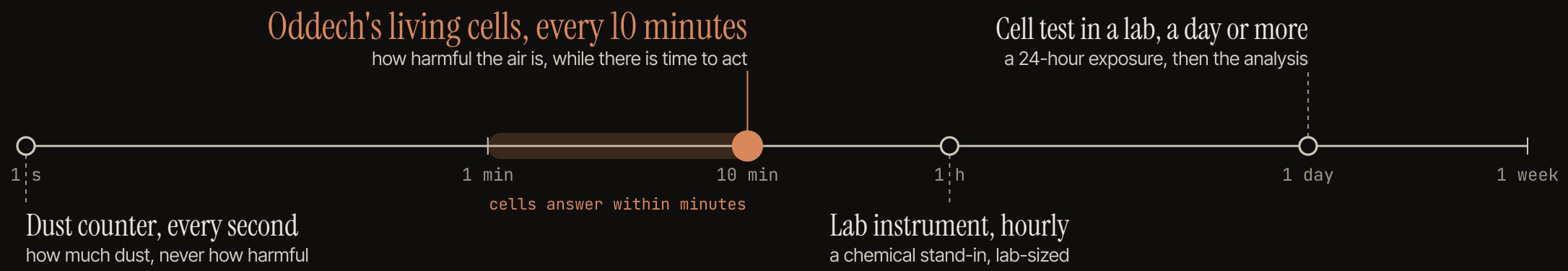


Same dust mass, different harm. The readout is simple: exposed wells over controls, where **1.0×** means clean air.

Every air monitor weighs the dust. *Oddech asks living cells how much it hurts.*

Oddech

Read in minutes, *for cents a reading.*



Within minutes

In lab studies, airway cells react to particles within minutes, far faster than classic toxicity tests.

Read alive

Live-cell readouts follow stress in real time; classic lab tests use the cells up.

About 2.5 cents a reading

One €25 cartridge gives about 1,000 readings a week.

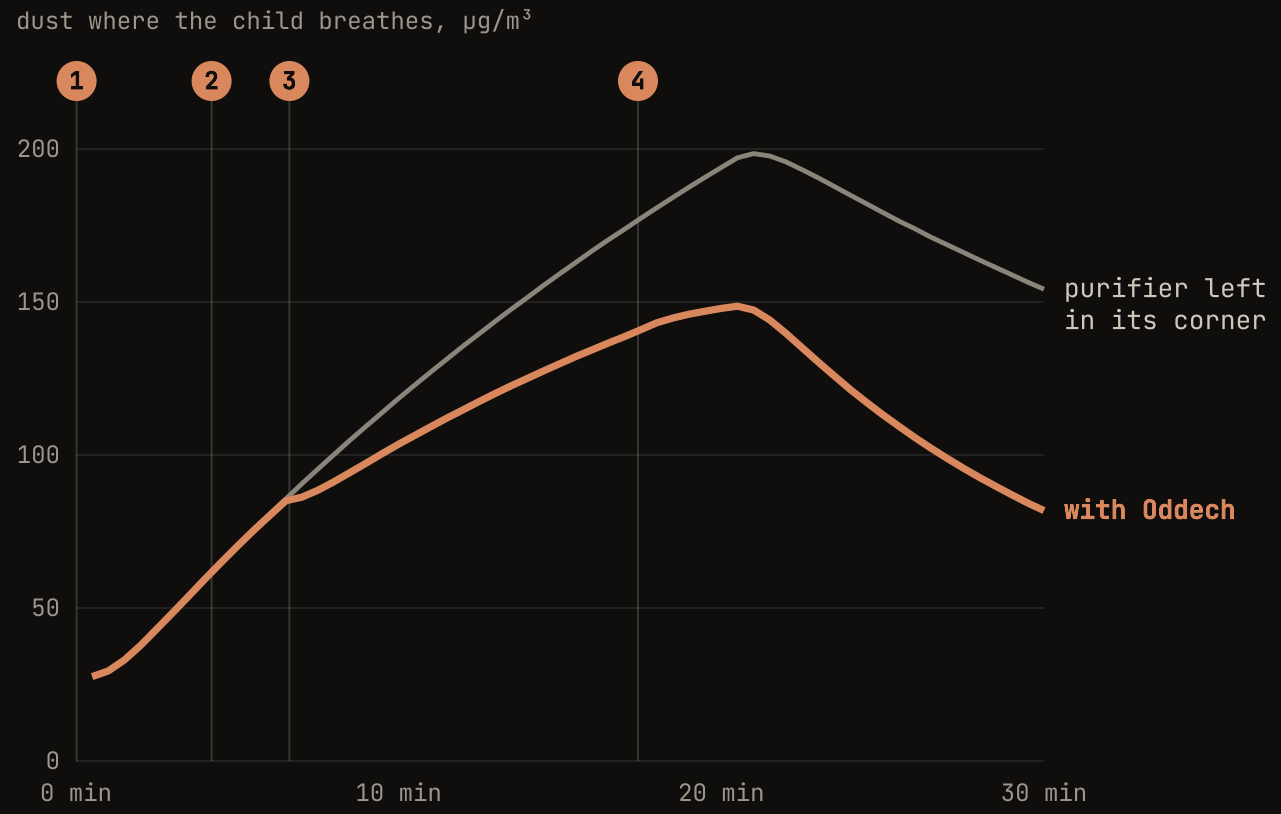
About 36 cents a child a day

A room of twenty, with the robot and a fresh cartridge every week.

Sources: Milici 2021; Corteselli 2018; Puthussery 2018; Huang 2021 · prices: planning assumptions, not quotes

Oddech

Same flat, same purifier: *the child breathes a fifth less dust.*



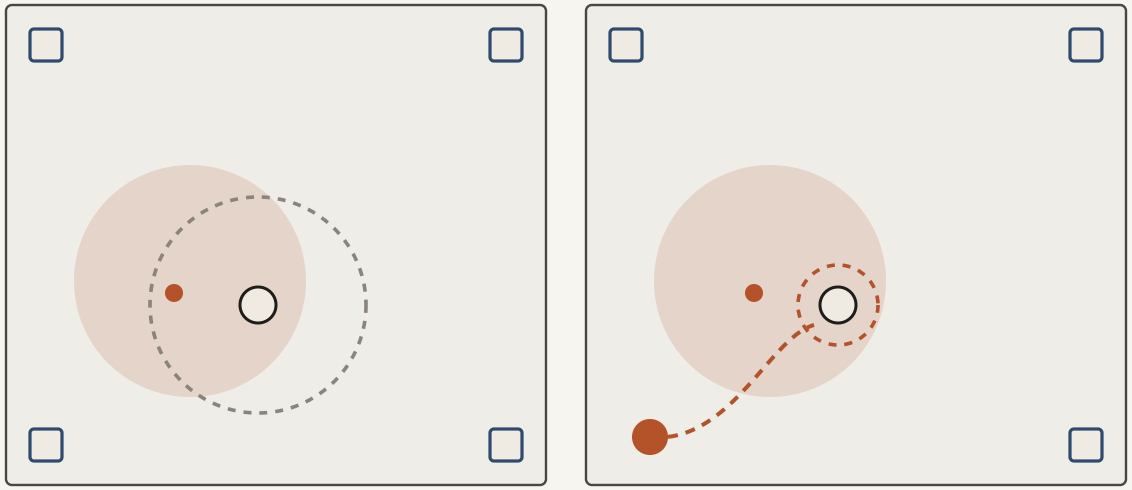
WHAT ODDECH DID, ONE EVENING

1. Switches the purifier on
2. Turns it up to high
3. Has it moved towards the play mat
4. Says when to open the window

Lower for the child in every model room we tried.

Oddech

One robot that moves *beats four sensors that stay put.*

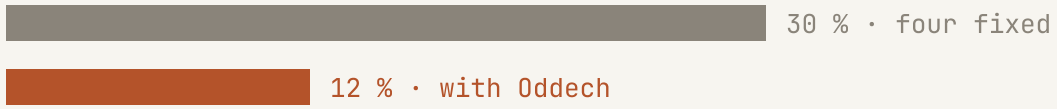


Four fixed sensors

Three fixed + Oddech

dashed ring: how unsure each is about the child's air

Typical error in the air people breathe



Less than half the error

in the air people breathe, in every model room.

It chooses where to go

It also beats a robot that wanders at random: the gain is in the choosing.

Living cells cannot sit in every corner

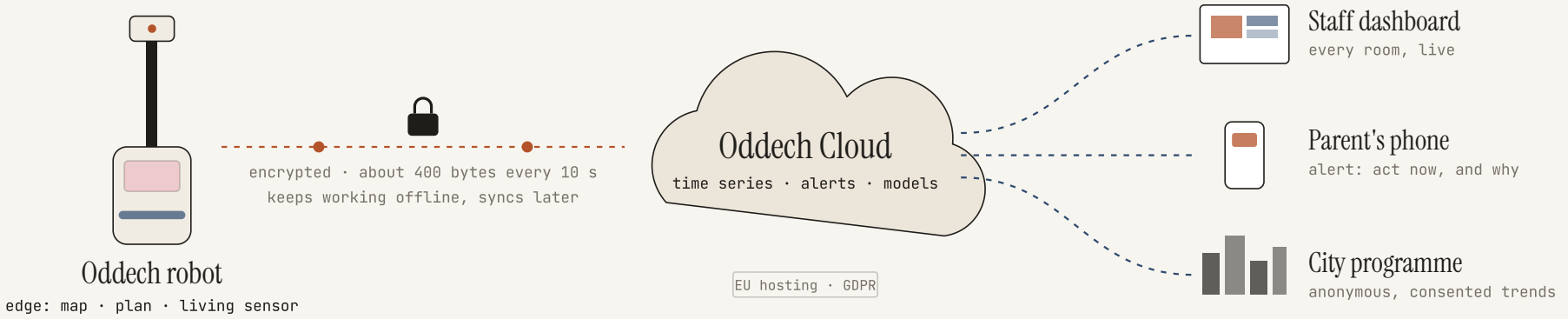
They need 37 °C, moist air and a fresh cartridge each week. One robot carries one cartridge everywhere.

One device to look after

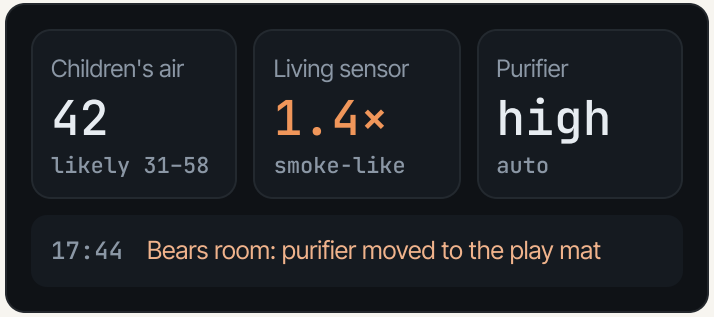
It works with the purifier you already have.

Oddech

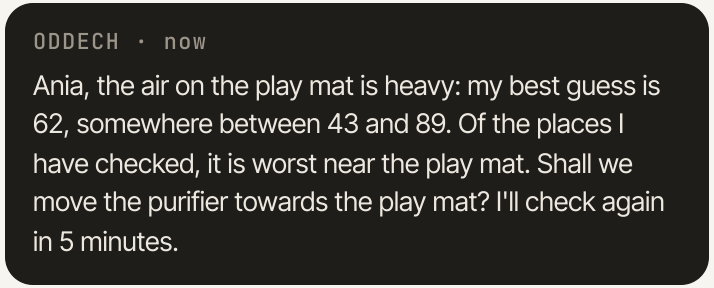
Each user gets *what they can act on.*



STAFF AND DIRECTOR · EVERY ROOM, LIVE



PARENTS · ON THE PHONE



CITY PROGRAMME · EVERY MONTH

Anonymous, consented trends

Which rooms breathe worst on smog evenings, district by district: where to fund purifiers, boiler swaps and pilots.

Phone message: Oddech's own words from the model evening · dashboard: working design on our site, demonstration values

Oddech

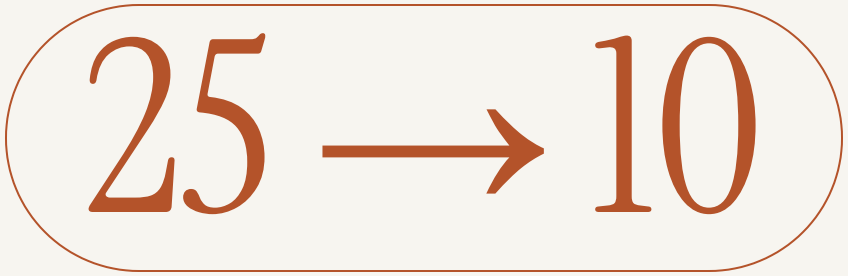
Start where the risk is highest *and someone can act.*

01 **Kindergartens** in Kraków: twenty children on the floor, one director deciding. Our first market.

02 **Care homes**: residents with lung disease, indoors all day.

03 **Parents** of children with asthma: a home station and the phone app.

04 **City programmes**: co-fund pilots, receive anonymous indoor data.



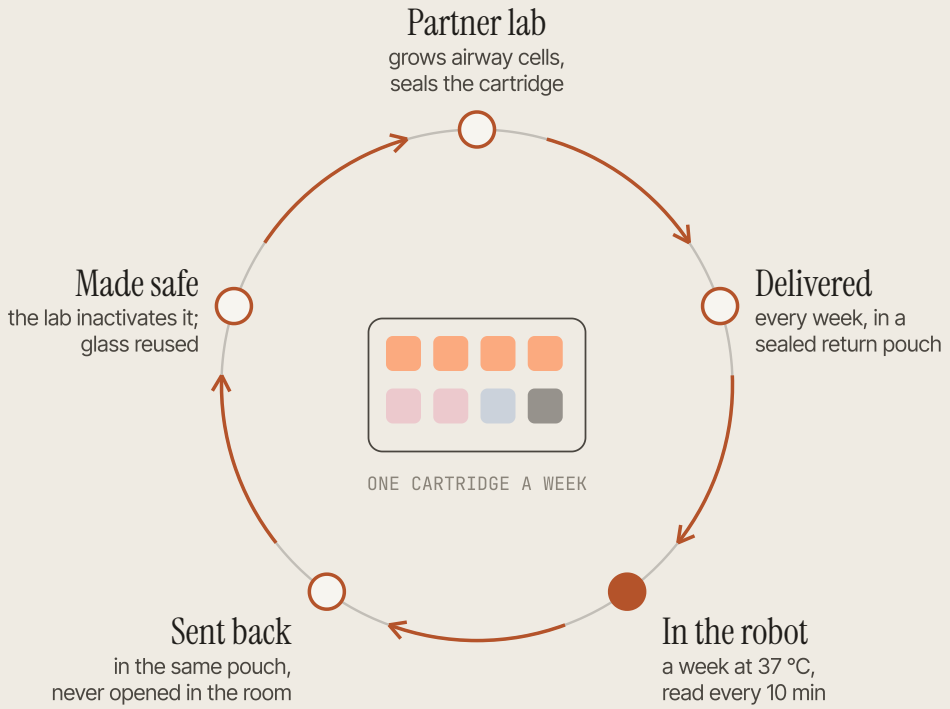
The EU limit for annual PM2.5 falls from 25 to 10 µg/m³ in 2030. Institutions will be asked what they did about the air inside.

PAYS	the kindergarten: one monthly fee per room
USES	staff and the director: live dashboard, alerts with an action
HEARS	parents: phone alerts and a monthly air report
GAINS	the children: cleaner air where they breathe

Source: Directive (EU) 2024/2881

Oddech

Rent the robot, *refill the living sensor.*



A **Oddech Room**: the robot as a service, one monthly fee per room.

B **Living cartridges**: a fresh one every week, returned like capsules.

C **Oddech Cloud**: a dashboard per site, an API for purifier makers.

D **Lab reports**: filter samples confirmed by a partner lab.

about 7 months to pay back a robot

Planning assumptions, not quotes · used cartridges handled as laboratory waste: WHO Laboratory Biosafety Manual 2020

Oddech

What could go wrong, *and our answer.*

Cells drift or tire	Controls, a lamp reference and a blank in every cartridge; a fresh cartridge each week.
Heat, dry air, condensation	A 37 °C bay, sample air kept moist, heated lines; a reading counts only if both stayed in range.
Biological waste	A sealed cartridge with its own filter, sent back in its pouch and made safe by the partner lab.
Safety around children	Slow near people, mast down while it drives, a soft bumper, no camera.
Trust	Every message says how sure it is; the wording is tested with people before a pilot.

PLAN B

Rooms evenly mixed: **Oddech Station**, a fixed living sensor.

Cells too fragile: the robot **collects filters** for a lab.

No one pays: **license** the robot's brain to purifier makers.

Sources: Juárez-Facio et al. 2024; Chidekel et al. 2012; WHO Laboratory Biosafety Manual 2020

Oddech

From Kraków *to a first room.*

14 Oct
Field visit, Kraków

16 Oct
Final presentation

27 Nov
Student Research Day, TUAT

26 Feb 2027
Entre Challenge

2027 · plan
Lab calibration, first pilot room

Step 1
Simulation
a known truth; this site

Step 2
Lab and chamber
calibrate the cells; a sealed room

Step 3
One real room
a kindergarten, with consent

Step 4
Pilot
several rooms, one smog season

Programme dates from the FL schedule; 2027 is our plan, not yet arranged

Oddech

Physics, robotics, biology, communication. *Take one away and the chain breaks.*

PHYSICS-INFORMED AI

Julian Evan Chrisnanto

TUAT, Japan · room map, robot's choices, dose

1

A room for a 2027 pilot

PHYSICAL AI AND ROBOTICS

Antonio Galiza Cerdeira Gonzalez, Ph.D.

Body, mast, motion and safety

2

A lab bench to calibrate the cells

RNA BIOINFORMATICS

Xiyue (Thea) Cao

TUAT, Japan · the Living Air Sensor

3

Introductions: city, purifier makers

HUMAN-MACHINE COMMUNICATION

Aleksandra Skrzypiec

Jagiellonian University · how Oddech speaks

4

Your honest view

Dziękujemy. Thank you. ありがとうございました。

Open the loop, *run it yourself.*

DEMO LINK

oddech-robot.streamlit.app

https://oddech-robot.streamlit.app/

Open the link, or scan the code with your phone: the room model, the living sensor, the 3D blueprint and the Gazebo twin.

SCAN TO OPEN

Not a stronger purifier: *knowing the air where the child breathes, and acting on it.*